CYBER THREAT MONITORING

LEVEL 1 - GUIDED CAPSTONE

Operation Night Ledger

A five-day, team-based blue-team project: build a monitored Windows lab, connect it to the Wazuh SIEM, investigate a simulated intrusion, and report your findings like a professional Security Operations Center.

Duration	5 days (Day 1 build -> Day 5 present)
Format	Group project, worked online from one shared folder
Tools	VirtualBox, Windows + Event Viewer, Sysmon, Wazuh
You play	The BlueHarbor Finance SOC (blue team)
All files	CAPSTONE_RESOURCES (this folder)

Start here: read 'How to use this guide', then begin Day 1.
Safety: run all scripts only inside your lab virtual machines.

Team members: _____ Cohort: _____

Contents

How to use this guide 3

1. Capstone overview 3

2. The scenario: Operation Night Ledger 4

Day 1 - Build the lab 5

Day 2 - Turn on Windows logging 7

Day 3 - Send the logs to Wazuh 8

Day 4 - The incident and investigation 10

Day 5 - Report and present 11

4. How you are graded 12

5. Troubleshooting 12

6. Glossary 13

7. Appendix - resource index 14

This guide is also a checklist: each day ends with a highlighted checkpoint box. Tick every item before moving on.

How to use this guide

Welcome to your **Cyber Threat Monitoring Level 1 Capstone**. Over five days your team will build a small security lab, switch on Windows logging, connect it to the **Wazuh** SIEM, investigate a simulated intrusion, and present your findings like a real Security Operations Center (SOC).

This is a **guided** capstone. Follow the days in order. Every step tells you exactly what to click or type, and every day ends with a **checkpoint** you can tick off.

> **One folder for everything.** All files you need are in the `CAPSTONE_RESOURCES` folder that came with this guide. Upload that one folder to your team's shared drive (Google Drive, OneDrive, or Teams). Nobody needs anything else.

> **Safety first.** Run every script and command only **inside your lab virtual machines** - never on your personal or work computer. The Day 4 "attack" is a harmless simulation. No real malware is used.

Legend used in this guide

DO THIS	A numbered action to perform.
CHECK	How to confirm the step worked.
TIP	Helpful advice or a shortcut.
WARNING	Something that can break your lab - read it.
FILE	A file to use from CAPSTONE_RESOURCES.

1. Capstone overview

What you will build

A two-machine lab inside **VirtualBox**:

- **CTM-WIN01** - a Windows workstation (the "victim"), with Windows auditing, **Sysmon**, and a **Wazuh agent**.
- **CTM-Wazuh** - the Wazuh all-in-one server (manager + indexer + dashboard), which collects and analyses the logs.

Learning outcomes

By the end, every team member can:

- Stand up a monitored Windows endpoint in a virtual lab.
- Enable and read the Windows logs a SOC relies on (Event Viewer).
- Collect endpoint logs centrally in a SIEM and write custom detection rules.
- Triage alerts, confirm them with raw evidence, and build an attack timeline.
- Map activity to **MITRE ATT&CK** and write an incident report.

Tools you will use (all free)

Tool	Role in the capstone
Oracle VirtualBox	Runs the virtual machines
Windows 11 / Server 2022 (eval)	The monitored endpoint
Windows Event Viewer	Read raw Windows logs (evidence)
Sysmon (Sysinternals)	Deep endpoint telemetry
Wazuh	SIEM - central collection, rules, dashboard
MITRE ATT&CK	Framework to classify attacker behaviour

The five days at a glance

Day	Theme	You finish with
Day 1	Build the lab	Two VMs on one network, snapshots taken
Day 2	Windows logging	Auditing + Sysmon on; Event Viewer views ready
Day 3	Wazuh SIEM	Agent ACTIVE; custom rules; logs flowing
Day 4	Attack & investigate	Triaged alerts; attack timeline; MITRE map
Day 5	Report & present	Incident report + team presentation

Team roles

Pick roles on Day 1 and record them in `07_Team_Templates / 01_Team_Charter.docx`. Teams of 3 combine roles; everyone investigates on Day 4.

Role	Owns
SOC Lead / Incident Commander	Stand-ups, task assignment, final timeline and decisions
Lab Engineer (Red Cell)	Builds and snapshots VMs; runs the Day 4 scenario
Windows Analyst	Event Viewer evidence and Windows Event IDs
Wazuh Analyst	Dashboard triage, custom rules, MITRE mapping
Scribe / Report Lead	Worksheet, incident report, presentation

2. The scenario: Operation Night Ledger

BlueHarbor Finance is a small (fictional) accounting firm. Your team is its SOC.

> At 07:45 this morning the Finance Manager reported that the file `Vendor_Bank_Details.csv` is missing from the shared finance workstation, and that "the computer felt slow last night." Helpdesk staff member **Jamie Santos (j.santos)** says they did not log in yesterday evening.

Management wants answers: Did someone break in, and how? What did they do? Was data taken or changed? Can they get back in? What must we fix? You will answer these on Days 4-5.

Day 1 - Build the lab

Goal: two virtual machines running and able to talk to each other on a private network. **Checkpoint:** you can ping between the VMs and open the Wazuh dashboard.

1.1 Download everything (do this first)

FILE

`02_Lab_Setup / Download_Links.md` has every link. Downloads total about 12 GB, so start them before anything else.

DO THIS

1. Download and install **VirtualBox 7.x** for your computer.
2. Download the **Wazuh OVA** (the ready-made SIEM virtual machine, ~5 GB).
3. Download a **Windows ISO**: Windows 11 Enterprise Evaluation (90 days). If your PC cannot run Windows 11, use Windows Server 2022 Evaluation instead.

> While downloads run, read `02_Lab_Setup / Lab_Network_Plan.md` and fill in the VM specs in your Team Charter.

1.2 Check your computer can run VMs

DO THIS

1. Reboot into your PC's BIOS/UEFI and make sure **virtualization** (Intel VT-x / AMD-V) is **enabled**. Without it, 64-bit VMs will not start.
2. Make sure you have at least **8 GB RAM** and **100 GB free disk**.

> If a member's laptop is too small, the team uses **one** strong PC as the "primary lab" and shares its screen during team calls. Decide this today.

1.3 Import the Wazuh server VM

DO THIS

1. VirtualBox menu: **File > Import Appliance**. Choose the Wazuh `.ova`.
2. Set RAM to **4096 MB** (8192 if you can), 2-4 CPUs. Name it **CTM-Wazuh**. Click Import.
3. Start the VM. At the login prompt sign in with **wazuh-user** / **wazuh**.
4. Type `ip -4 addr` and note the address (you will set this up in step 1.5).

1.4 Create the Windows endpoint VM

DO THIS

1. **Machine > New**. Name **CTM-WIN01**, Type Microsoft Windows, Version **Windows 11 (64-bit)**. VirtualBox 7 enables EFI, Secure Boot and TPM for you.
2. RAM **4096 MB**, CPUs **2**, disk **64 GB**. Attach the Windows ISO.
3. Install Windows. Create a local admin user named **ctmadmin** with a password you choose (write it

in your Team Charter). Set the computer name to **WIN-CTM-01**.

> To skip the "Microsoft account" requirement during Windows 11 setup, choose the **Windows Server** option instead, or select a limited-setup/"domain join" path when offered. A local account is all the lab needs.

1.5 Put both VMs on one private network

You need a **Host-only** network so the two VMs can talk privately, plus **NAT** so each can reach the internet for updates.

DO THIS

1. VirtualBox: **Tools > Network > Host-only Networks > Create**. Leave the default `192.168.56.0/24` with DHCP enabled.
2. For **each** VM: **Settings > Network**. Adapter 1 = **NAT**. Adapter 2 = **Host-only Adapter** (select the one you just created). Enable both.
3. Start both VMs. Find each host-only IP (starts with 192.168.56): on Wazuh type `ip -4 addr`; on Windows open Command Prompt and type `ipconfig`.
4. Record both IPs in your Team Charter.

CHECK

1. From the Windows VM, open a browser and go to `https://<WAZUH-IP>`. Accept the certificate warning. Log in with **admin** / **admin**. You should see the Wazuh dashboard. In the dashboard menu, open **About** and record the **Wazuh version** - you need it on Day 3.

1.6 Take your first snapshots

> Snapshots are your safety net. If anything breaks later, you roll back in seconds instead of rebuilding.

DO THIS

1. For each VM: VirtualBox **> (select VM) > Snapshots > Take**.
2. Name them **Day1-Clean-Build**.

> **Day 1 checkpoint - all must be true**
>
> ```
> [ ] VirtualBox installed, virtualization enabled
> [ ] CTM-Wazuh and CTM-WIN01 both boot
> [ ] Both VMs have a 192.168.56.x address
> [ ] Wazuh dashboard opens from the Windows VM; version recorded
> [ ] "Day1-Clean-Build" snapshot taken on both VMs
> [ ] Team Charter filled in (roles, IPs, version)
> ```

Day 2 - Turn on Windows logging

Goal: make the Windows endpoint record the security events a SOC needs. **Checkpoint:** `Preflight_Check.ps1` shows the logging items as PASS.

> **Running the scripts.** All Day 2-4 scripts run inside the Windows VM. Right-click **Windows PowerShell > Run as administrator**. If scripts are blocked, run this once in that same window:
> `Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass`
> Then run a script with `.\ScriptName.ps1`. Copy the CAPSTONE_RESOURCES folder into the VM first (shared folder, or just download it from your shared drive inside the VM).

2.1 Create the company environment

FILE

`03_Windows_Logging / Seed_Lab_Environment.ps1`

This creates three staff accounts (**j.santos**, **m.reyes**, **a.cruz**) and the `C:\CapstoneFinance` folder with sample business files. Note: j.santos is given a deliberately **weak password** - that is part of the story.

DO THIS - run it, then confirm the users appear with `Get-LocalUser`.

2.2 Enable Windows audit logging

FILE

`03_Windows_Logging / Enable_Audit_Logging.ps1`

This turns on auditing for logons, account changes, process creation and more; adds the **command line** to process events; enables **PowerShell script block logging**; and makes the logs bigger so evidence is not overwritten.

DO THIS - run it. It prints the audit policy at the end; screenshot that.

> After this runs, these events start recording: 4624/4625 (logon success/failure), 4720/4732 (account created / added to group), 4688 (process with command line), 4698 (scheduled task), 4104 (PowerShell code). See the cheat sheet below.

2.3 Install Sysmon

FILE

`03_Windows_Logging / Install_Sysmon.ps1` and `sysmon-capstone.xml`

Sysmon adds deep detail: every process (with hashes and parent), network connections, file creation/deletion, registry autostart changes and DNS queries. The script downloads Sysmon from Microsoft and installs it with our tuned configuration.

DO THIS - run it.

CHECK - open Event Viewer > Applications and Services Logs > Microsoft > Windows > Sysmon > Operational. You should see events appearing.

2.4 Import the ready-made Event Viewer views

FILE

`03_Windows_Logging / EventViewer_Custom_Views\` (four XML files)

These give you one-click filtered views instead of scrolling through thousands of events.

DO THIS

1. Open **Event Viewer** (press Start, type `eventvwr.msc`).
2. Right-click **Custom Views > Import Custom View...**
3. Import each of the four files: **CV1 Authentication**, **CV2 Account Changes**, **CV3 Persistence and Execution**, **CV4 PowerShell and Defender**.

2.5 Learn the key Event IDs

Keep `03_Windows_Logging / Windows_Event_ID_Cheat_Sheet.md` open while you work. The essentials:

Event ID	Meaning	Why it matters
4625	Failed logon	Password guessing / brute force
4624	Successful logon	Who got in, and how (Logon Type)
4720	User account created	Attacker making a backdoor account
4732	Added to a local group	Privilege escalation (e.g. Administrators)
4688	New process	What was run (with command line)
4698	Scheduled task created	Persistence
4104	PowerShell script block	Shows the actual code - even decoded
1102	Security log cleared	Always suspicious

> **Day 2 checkpoint**
> `[ ] Seed users and C:\CapstoneFinance created`
> `[ ] Audit logging enabled (screenshot of auditpol output)`
> `[ ] Sysmon service running and logging events`
> `[ ] Four custom views imported into Event Viewer`
> `[ ] Preflight_Check.ps1 shows logging items PASS`
> `[ ] Snapshot taken: "Day2-Logging-On"`

Day 3 - Send the logs to Wazuh (SIEM)

Goal: the Windows endpoint ships its logs to Wazuh, and your custom rules raise alerts. **Checkpoint:** the agent shows **ACTIVE** and your test events create alerts.

3.1 Install the Wazuh agent on Windows

FILE

`04_Wazuh / Install_Wazuh_Agent.ps1`

This installs the agent, points it at your Wazuh server, and adds the extra log sources (Sysmon,

PowerShell, Defender) plus **File Integrity Monitoring** on the finance folder.

> The agent version must be the **same or older** than your Wazuh server version (the one you recorded on Day 1). Pass it with `-Version`.

DO THIS - in the Windows VM, run (use your own IP and version):

```
.\Install_Wazuh_Agent.ps1 -WazuhIP 192.168.56.101 -Version 4.9.2
```

CHECK

1. The script prints the agent log; look for **"Connected to the server"**.
2. In the Wazuh dashboard open **Agents**. **WIN-CTM-01** should be **ACTIVE** (green).

> If the agent is not active: check the IP, confirm both VMs are on the host-only network, and that ports 1514/1515 are reachable (run `Preflight_Check.ps1 -WazuhIP <ip>`). See Troubleshooting.

3.2 Add the custom detection rules

FILE

`04_Wazuh / local_rules.xml`

These rules (IDs 100200-100210) are the brain of your SOC. They turn raw events into meaningful alerts: brute force, new admin accounts, scheduled tasks, encoded PowerShell, registry persistence, finance-file tampering, and Defender detections - each mapped to MITRE.

DO THIS

1. In the dashboard: **menu > Server management > Rules**.
2. Click **Manage rules files**, then **Custom rules**, and edit **local_rules.xml**.
3. Keep anything already there. Paste the contents of our `local_rules.xml` below it. Click **Save**.
4. When prompted, **Restart** the manager (or on the server console run `sudo systemctl restart wazuh-manager`).

> Alternative on the server console: edit `/var/ossec/etc/rules/local_rules.xml`, then restart the manager. After a restart, check `sudo tail -n 50 /var/ossec/logs/ossec.log` for rule errors.

3.3 Prove it works (smoke test)

DO THIS - in the Windows VM, generate a couple of safe test events:

1. Fail a logon on purpose a few times: open PowerShell and run `runas /user:WIN-CTM-01\j.santos cmd` and type a wrong password 3-4 times.
2. Create and remove a test file in the watched folder: `"test" | Out-File C:\CapstoneFinance\test.txt ; Remove-Item C:\CapstoneFinance\test.txt`

CHECK - in the dashboard go to Threat Hunting (or Discover) and search `rule.groups:"ctm_capstone"`. You should see your alerts. Use the queries in `04_Wazuh / Wazuh_Dashboard_Queries.md`.

Day 3 checkpoint

```
[ ] Wazuh agent ACTIVE in the dashboard
[ ] Sysmon / PowerShell / Defender channels + FIM configured
[ ] Custom rules loaded; manager restarted without errors
[ ] Smoke-test alerts visible under rule.groups:"ctm_capstone"
[ ] Snapshot taken on both VMs: "Day3-Wazuh-Ready"
```

Day 4 - The incident and your investigation

Goal: detect, investigate and document the intrusion. **Checkpoint:** a complete attack timeline and all alerts triaged.

4.1 Run the scenario (Red Cell only)

The attack is launched by a script (`Run_Scenario.ps1`) that your **instructor provides to the Red Cell** (the Lab Engineer). It is deliberately NOT in the student pack, so the rest of the team investigates blind, like a real SOC. The Red Cell runs it inside the Windows VM as Administrator, does **not** read it, and does **not** share its output. First take a snapshot named **Day4-Before-Attack**.

DO THIS (Red Cell): run the script your instructor provided. It takes about 6 minutes and prints only "Stage X of 8 complete". Then wait 2-3 minutes for Wazuh to index events.

Plan B. If your live lab is broken (agent never connected, a VM died), ask your instructor for the **backup dataset** - the same incident exported as CSV and JSON files you can open in Excel and a text editor. Your investigation and report are identical.

4.2 How to investigate (the SOC method)

Work this loop for every alert. Record everything in `07_Team_Templates / 03_Alert_Triage_Tracker.xlsx` and `04_Investigation_Worksheet.docx`.

1. **Spot** - in Wazuh, search `rule.groups:"ctm_capstone"` and sort by time.
2. **Understand** - open the alert; read rule.description, rule.level, MITRE ID, and the event fields (account, command line, file).
3. **Confirm** - find the **same event in Event Viewer** using your custom views. This raw evidence is what goes in the report.
4. **Record** - add a row to the triage tracker: time, rule.id, Event ID, true/false positive, severity, MITRE, screenshot name.
5. **Connect** - place it on the timeline. Ask: what came before, what came after?

Take a screenshot of **every** confirmed alert (both the Wazuh view and the Event Viewer entry). Name them like `T03_4720_new_user.png`. Screenshots are your evidence and your presentation material.

4.3 What to look for (without spoiling it)

The intrusion moves through these ATT&CK stages. Find the evidence for each:

Stage	Question to answer	Where to look
Initial Access	How did they get in? Whose account?	4625, 4624; rule 100201
Discovery	What did they look at?	Sysmon 1 / 4688; rule 100206
Privilege / Persistence	New account? Auto-start? Task?	4720, 4732, 4698, Sysmon 13; rules 100202-100207
Execution	What code ran, and was it hidden?	4104, Sysmon 1; rule 100205
Collection / Exfil	What data was staged, changed, deleted, sent out?	FIM, Sysmon 11/22/3/26; rules 100208/100210
Tooling	Did AV catch anything?	Defender 1116/1117; rule 100209

Answer the 15 questions in the Investigation Worksheet. Decode the encoded PowerShell with **CyberChef** (From Base64) or by reading Event **4104**, which logs the decoded code.

> **Day 4 checkpoint**
> `[ ] Every ctm_capstone alert triaged in the tracker`
> `[ ] Each confirmed with an Event Viewer screenshot`
> `[ ] All 15 worksheet questions answered with evidence`
> `[ ] Attack timeline drafted (minute by minute)`
> `[ ] Each stage mapped to MITRE ATT&CK`

Day 5 - Report and present

Goal: deliver a professional incident report and a team presentation. **Checkpoint:** both complete and saved in the shared folder.

5.1 Write the incident report

FILE

`07_Team_Templates / 05_Incident_Report_Template.docx`

Fill every section. The executive summary must be readable by a non-technical manager. Your timeline and MITRE map come straight from Day 4. End with a clear **root cause** (hint: how strong was j.santos's password, and what rights did the account have?) and at least **five prioritised recommendations**.

5.2 Clean up and recover

After all evidence is collected, the Red Cell runs the **cleanup script your instructor provided** (`Cleanup_Scenario.ps1`) to remove the backdoor account, scheduled task, Run key and dropped files, and to restore the finance data. In your report, describe this as your **containment, eradication and recovery**. (A real team would also reset the weak password and remove unnecessary admin rights - list that too.)

5.3 Build the presentation

FILE

`07_Team_Templates / 06_Final_Presentation_Outline.docx`

10-12 minutes, **every member speaks**. Lead with the story (the timeline slide is your strongest). Show the live Wazuh dashboard or your screenshots.

5.4 Peer evaluation

Each member completes `08_Assessment / Peer_Evaluation_Form.docx` and submits it privately to the instructor.

> **Day 5 checkpoint (final hand-in)**
> `[ ] Incident report complete`
> `[ ] Presentation delivered; every member spoke`
> `[ ] Triage tracker, worksheet, charter, stand-up log saved`
> `[ ] Lab cleaned up / recovered; final snapshot taken`
> `[ ] Peer evaluations submitted`

4. How you are graded

See `08_Assessment / Grading_Rubric.xlsx` for the full breakdown (100 points). Headline weights:

Area	Weight
Lab build & logging (Days 1-2)	15
Wazuh integration (Day 3)	15
Detection quality	10
Investigation & triage (Day 4)	20
Timeline & MITRE mapping	10
Incident report	15
Final presentation	10
Teamwork & documentation	5

5. Troubleshooting

Problem	Try this
VM won't start / 64-bit greyed out	Enable Intel VT-x / AMD-V in BIOS. Disable Hyper-V on the host if it conflicts.
VMs can't reach each other	Both need Adapter 2 = the **same** Host-only Adapter. Check IPs are both 192.168.56.x.
Scripts are blocked	Run `Set-ExecutionPolicy -Scope Process -ExecutionPolicy Bypass` in that PowerShell window, as Administrator.
Wazuh agent not ACTIVE	Run `Preflight_Check.ps1 -WazuhIP <ip>`. Confirm ports 1514/1515 reachable; restart `WazuhSvc`; read `ossec.log`.
Agent version error	Agent version must be <= server version. Reinstall with the right `-Version`.
Custom rules not firing	Restart the manager. Check `/var/ossec/logs/ossec.log` for XML errors. Test with `wazuh-logtest`.
No Sysmon events	Confirm the **Sysmon64** service is running; re-run `Install_Sysmon.ps1`.
Dashboard won't load	Give the Wazuh VM more RAM (8 GB). Wait 2-3 minutes after boot. Use `https://` and accept the certificate.
Live lab broke on Day 4	Ask your instructor for the backup dataset and continue the investigation from the files.

Still stuck after 15 minutes? Post in your team channel with a screenshot of the error and what you already tried.

6. Glossary

Term	Plain meaning
SIEM	Security Information and Event Management - software that collects logs and raises alerts (Wazuh).
SOC	Security Operations Center - the team that watches for and responds to attacks (you).
Agent	Small program on a computer that sends its logs to the SIEM.
Sysmon	Microsoft tool that records detailed endpoint activity.
Event ID	A number Windows uses to label a type of event (e.g. 4625 = failed logon).
FIM	File Integrity Monitoring - alerts when watched files change.
Triage	Quickly deciding if an alert is real and how urgent it is.
True/False positive	A real detection vs. a harmless one.
Persistence	Techniques that let an attacker regain access after a reboot.
MITRE ATT&CK	A public catalogue of attacker techniques, each with an ID like T1110.
Snapshot	A saved state of a VM you can instantly roll back to.

7. Appendix - resource index

Everything below is in the single **CAPSTONE_RESOURCES** folder.

Folder	Used on	Contents
01_Guide	All	This guide (PDF)
02_Lab_Setup	Day 1	Download links, network plan, Preflight_Check.ps1
03_Windows_Logging	Day 2	Seed, audit, Sysmon scripts + config, custom views, cheat sheet
04_Wazuh	Day 3	Agent installer, ossec additions, local_rules.xml, dashboard queries
05_Attack_Simulation	Day 4	Scenario brief (attack & cleanup scripts are provided by your instructor)
07_Team_Templates	All	Charter, stand-ups, triage tracker, worksheet, report, presentation
08_Assessment	Day 5	Grading rubric, peer evaluation

End of guide - good luck, SOC team. Work the evidence, trust the timeline, and tell the story.